Zhihui Tian

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Summary

Ph.D. candidate in Electrical and Computer Engineering at the University of Florida developing scalable machine learning methods for large-scale 3D microstructure evolution. Combines graph neural networks and physics-regularized modeling to improve the efficiency and long-term stability of mesoscale microstructure evolution prediction. Seeking industry research / ML roles and postdoctoral positions.

Education

Ph.D. **University of Florida**, Electrical and Computer Engineering — Gainesville, FL; May 2023 – present

- Advised by Prof. Joel B. Harley (SmartDATA Lab).
- Research on scalable, physics-guided machine learning for 3D microstructure evolution.

M.S. **University of Florida**, Electrical and Computer Engineering — Gainesville, FL; Aug 2021 – May 2023

Experience

Lawrence Livermore National Laboratory (LLNL), Collaborative Research Project — Remote / Gainesville, FL; Sept 2025 – present; 1 year 1 month

Scalable hybrid CNN-GNN surrogate for 3D grain growth.

- Designed a hybrid bijective autoencoder + graph neural network surrogate that evolves 3D grain microstructures in a compressed latent space.
- Developed a latent-space inference strategy achieving up to 117x memory reduction and 115x runtime improvement on 3D meshes up to 160^3 relative to a GNN-only baseline.
- Work published in Acta Materialia (2026) with collaborators at LLNL.

SmartDATA Lab, University of Florida, Graduate Research Assistant — Gainesville, FL; May 2023 – present; 3 years 5 months

- Physics-regulated interpretable ML for 3D microstructure evolution: built a scalable 3D architecture with spatiotemporal extrapolation and inclination/misorientation-dependent grain boundary dynamics; applied it to Lab-DCT experimental data.
- Computer vision for experimental microstructure analysis: built a TEM image preprocessing and registration pipeline; fine-tuned U-Net and the Segment Anything Model for microstructure segmentation.
- Gaussian-sampled mode filter for grain growth: developed Gaussian-sampled kernels that emulate grain boundary motion, mitigate lattice pinning, and improve adherence to the von Neumann-Mullins law over Monte Carlo Potts models.

Publications

Scaling kinetic Monte-Carlo simulations of grain growth with combined convolutional and graph neural networks
Zhihui Tian, Ethan Suwandi, Tomas Oppelstrup, Vasily V. Bulatov, Joel B. Harley, Fei Zhou
doi.org/10.1016/j.actamat.2026.122153 (Acta Materialia, 2026)

A mode filter approach to grain growth - improved performance in lattice pinning and von Neumann-Mullins relation
Zhihui Tian, Lin Yang, Kang Yang, Vishal Yadav, Amanda Krause, Michael Tonks, Joel B. Harley
(Acta Materialia (under review), 2025)

Improved PCA reconstruction-based unsupervised anomaly detection in uncontrolled structural health monitoring with correntropy
Kang Yang, Zhenhan Lin, Zekun Yang, Zhihui Tian, Jie Ma, Jose C. Principe, Joel B. Harley
ieeexplore.ieee.org/document/11078390 (IEEE Transactions on Industrial Informatics, 2025)

Weight decay optimized unsupervised autoencoder-based anomaly detection in uncontrolled dynamic structural health monitoring
Kang Yang, Zekun Yang, Zhihui Tian, Joel B. Harley
doi.org/10.1007/978-3-031-94895-4_3 (Dynamic Data Driven Applications Systems (DDDAS), Lecture Notes in Computer Science, Springer, 2026)

Quantifying heterogeneous ecosystem services with multi-label soft classification
Zhihui Tian, Jeremy Upchurch, Gabriel A. Simon, Jose Dubeux, Alina Zare, Chang Zhao, Joel B. Harley
doi.org/10.1109/IGARSS53475.2024.10642804 (IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2024)

Skills

Machine Learning
Graph neural networks, convolutional autoencoders, physics-informed / physics-regularized models, surrogate modeling, unsupervised anomaly detection

Computer Vision
Image segmentation (U-Net, Segment Anything Model), image registration and preprocessing, TEM and Lab-DCT microstructure data

Materials Simulation
Grain growth, kinetic Monte Carlo Potts models, mode filter models, 3D microstructure evolution, von Neumann-Mullins analysis

Programming and Tools
Python, PyTorch, PyTorch Geometric, NumPy / SciPy, MATLAB, Git, Linux, HPC clusters

Languages

Chinese (Mandarin)
Native speaker

English
Fluent